

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

DATA PRE-PROCESSING

```
In [3]: df = pd.read_csv('mymoviedb.csv',lineterminator = '\n')
```

```
In [4]: df.head()
```

Out[4]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	
0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	A
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	A
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	A

In [5]: `df.tail()`

Out[5]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language
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9822	1973-10-15	Badlands	A dramatization of the Starkweather-Fugate killing.	13.357	896	7.6	
9823	2020-10-01	Violent Delights	A female vampire falls in love with a man she ...	13.356	8	3.5	
9824	2016-05-06	The Offering	When young and successful reporter Jamie finds...	13.355	94	5.0	
9825	2021-03-31	The United States vs. Billie Holiday	Billie Holiday spent much of her career being ...	13.354	152	6.7	
9826	1984-09-23	Threads	Documentary style account of a nuclear holocau...	13.354	186	7.8	



In [6]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
Release_Date      9827 non-null object
Title             9827 non-null object
Overview          9827 non-null object
Popularity        9827 non-null float64
Vote_Count        9827 non-null int64
Vote_Average      9827 non-null float64
Original_Language 9827 non-null object
Genre             9827 non-null object
Poster_Url        9827 non-null object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

```
In [7]: df['Genre']
```

```
Out[7]: 0      Action, Adventure, Science Fiction
      1      Crime, Mystery, Thriller
      2      Thriller
      3      Animation, Comedy, Family, Fantasy
      4      Action, Adventure, Thriller, War
      ...
      9822      Drama, Crime
      9823      Horror
      9824      Mystery, Thriller, Horror
      9825      Music, Drama, History
      9826      War, Drama, Science Fiction
      Name: Genre, Length: 9827, dtype: object
```

```
In [8]: df['Genre'].head()
```

```
Out[8]: 0      Action, Adventure, Science Fiction
      1      Crime, Mystery, Thriller
      2      Thriller
      3      Animation, Comedy, Family, Fantasy
      4      Action, Adventure, Thriller, War
      Name: Genre, dtype: object
```

```
In [9]: df.duplicated().sum()
```

```
Out[9]: 0
```

```
In [10]: df.describe()
```

```
Out[10]:
```

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

Exploration Summary

1. we have a dataframe consisting of 9827 rows and 9 columns.
2. our dataset looks a bit tidy with no NaNs nor duplicated values.
3. Release_Date column needs to be casted into date time and to extract only the year value.
4. Overview , Original_Language and Poster-Url wouldn't be so useful during analysis , so we will drop them.
5. there is noticeable outliers in popularity column
6. Vote_Average better be categorised for proper analysis.

7. Genre column has coma separated values and white spaces that needs to be handled and casted into category.

In [11]: `df.head()`

Out[11]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	
0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	A
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	A
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	A

In [12]: `df['Release_Date'] = pd.to_datetime(df['Release_Date'])`
`print(df['Release_Date'].dtypes)`
 datetime64[ns]

In [13]: `df['Release_Date'] = df['Release_Date'].dt.year`
`df['Release_Date'].dtypes`

Out[13]: dtype('int64')

In [14]: `df.head()`

Out[14]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	
0	2021	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	A
1	2022	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	
2	2022	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	
3	2021	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	A
4	2021	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	A



Dropping the Columns

In [15]: `cols = ['Overview' , 'Original_Language' , 'Poster_Url']`

In [16]: `df.drop(cols , axis = 1 , inplace = True)`
`df.columns`

Out[16]: `Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average', 'Genre'], dtype='object')`

In [17]: `df.head()`

Out[17]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	8.1	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	6.3	Thriller
3	2021	Encanto	2402.201	5076	7.7	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	7.0	Action, Adventure, Thriller, War

categorizing Vote_Average column

we would cut the Vote_Average values and make 4 categories : **Popular** , **average** , **below_Avg** , **not_popular** to describe it more using `categorize_col()` function provided above.

In [18]:

```
#user-define function
def categorize_col(df , col , labels):

    edges = [df[col].describe()['min'],
             df[col].describe()['25%'],
             df[col].describe()['50%'],
             df[col].describe()['75%'],
             df[col].describe()['max']]

    df[col] = pd.cut(df[col] , edges , labels = labels , duplicates = 'drop')
    return df
```

In [20]:

```
labels = ['not_popular' , 'below_avg' , 'average' , 'popular']
categorize_col(df , 'Vote_Average' , labels)
df['Vote_Average'].unique()
```

Out[20]: [popular, below_avg, average, not_popular, NaN]
Categories (4, object): [not_popular < below_avg < average < popular]

In [21]: `df.head()`

Out[21]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	below_avg	Thriller
3	2021	Encanto	2402.201	5076	popular	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	average	Action, Adventure, Thriller, War

```
In [23]: df['Vote_Average'].value_counts()
```

```
Out[23]: not_popular    2467
         popular      2450
         average      2412
         below_avg    2398
         Name: Vote_Average, dtype: int64
```

```
In [25]: df.dropna(inplace = True)

         df.isna().sum()
```

```
Out[25]: Release_Date    0
         Title           0
         Popularity      0
         Vote_Count      0
         Vote_Average    0
         Genre           0
         dtype: int64
```

```
In [26]: df.head()
```

```
Out[26]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	below_avg	Thriller
3	2021	Encanto	2402.201	5076	popular	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	average	Action, Adventure, Thriller, War

we would split the genres into a list and then explode our dataframe to have only one genre per row for each movie.

```
In [27]: df['Genre'] = df['Genre'].str.split(', ')

         df = df.explode('Genre').reset_index(drop = True)

         df.head()
```

```
Out[27]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

```
In [29]: #casting column into category

df['Genre'] = df['Genre'].astype('category')

df['Genre'].dtypes
```

```
Out[29]: CategoricalDtype(categories=['Action', 'Adventure', 'Animation', 'Comedy', 'Crime',
                                     'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
                                     'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
                                     'TV Movie', 'Thriller', 'War', 'Western'],
                                     ordered=False)
```

```
In [30]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
Release_Date    25552 non-null int64
Title           25552 non-null object
Popularity       25552 non-null float64
Vote_Count      25552 non-null int64
Vote_Average    25552 non-null category
Genre           25552 non-null category
dtypes: category(2), float64(1), int64(2), object(1)
memory usage: 849.5+ KB
```

```
In [31]: df.nunique()
```

```
Out[31]: Release_Date    100
Title                 9415
Popularity            8088
Vote_Count           3265
Vote_Average           4
Genre                  19
dtype: int64
```

```
In [32]: df.head()
```

```
Out[32]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

DATA VISUALIZATION

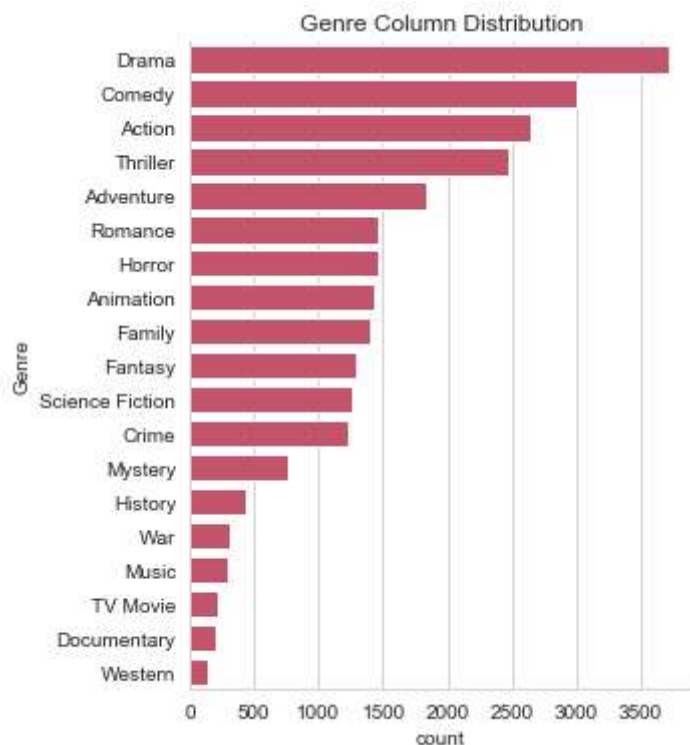

```
In [33]: sns.set_style('whitegrid')
```

1. What is the most frequent genre of movies released on Netflix ?

```
In [34]: df['Genre'].describe()
```

```
Out[34]: count      25552  
unique         19  
top            Drama  
freq           3715  
Name: Genre, dtype: object
```

```
In [46]: sns.catplot(y = 'Genre' , data = df , kind = 'count', order = df['Genre'].value_counts()  
plt.title('Genre Column Distribution')  
plt.show()
```



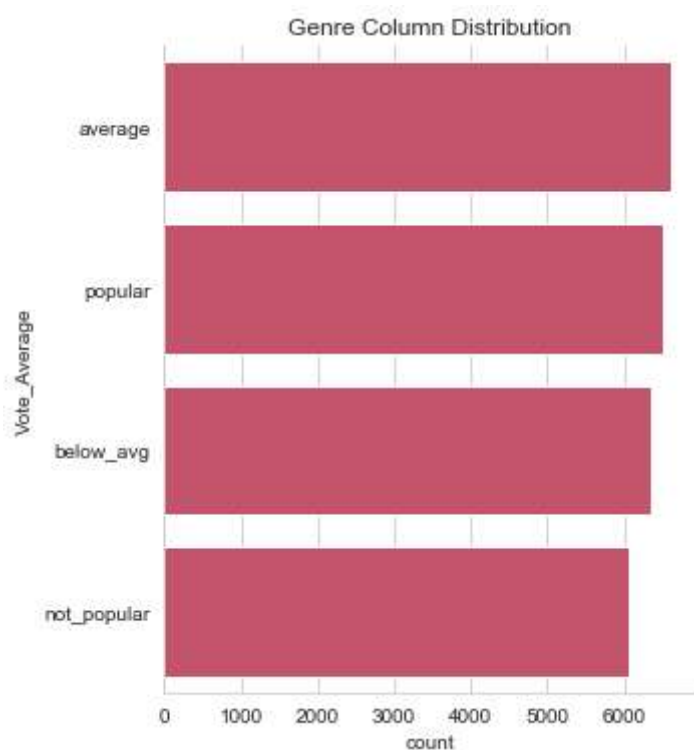
2. Which has highest votes in vote avg column?

In [36]: `df.head()`

Out[36]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

In [45]: `sns.catplot(y = 'Vote_Average', data = df , kind = 'count', order = df['Vote_Average'].unique())`
`plt.title('Genre Column Distribution')`
`plt.show()`



3. which movie got the highest popularity? what's it's genre?

In [40]: `df.head(2)`

Out[40]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure

```
In [41]: df[df['Popularity'] == df['Popularity'].max()]
```

```
Out[41]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction

4. which movie got the lowest popularity? what's its genre?

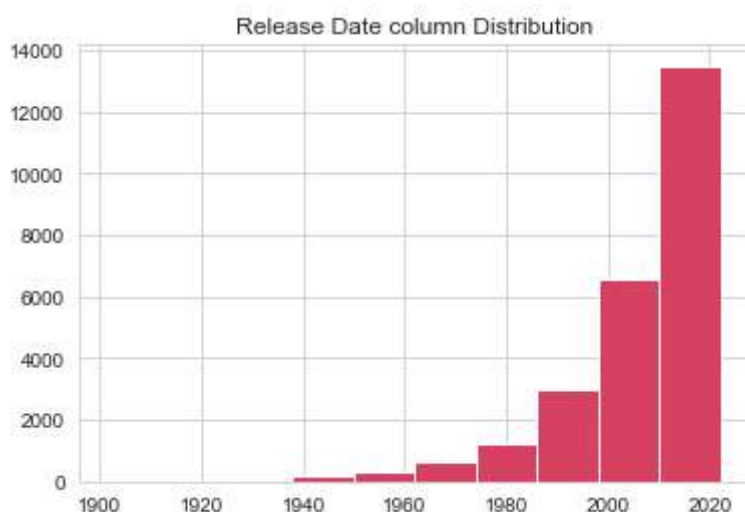
```
In [42]: df[df['Popularity'] == df['Popularity'].min()]
```

```
Out[42]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
25546	2021	The United States vs. Billie Holiday	13.354	152	average	Music
25547	2021	The United States vs. Billie Holiday	13.354	152	average	Drama
25548	2021	The United States vs. Billie Holiday	13.354	152	average	History
25549	1984	Threads	13.354	186	popular	War
25550	1984	Threads	13.354	186	popular	Drama
25551	1984	Threads	13.354	186	popular	Science Fiction

5. Which year has the most filmed movies?

```
In [48]: df['Release_Date'].hist(color = '#d64161')
plt.title("Release Date column Distribution")
plt.show()
```



Conclusion

Q1: What is the most frequent genre in the dataset? Drama genre is the most frequent genre in our dataset and has appeared more than 14% of the times among 19 other genres.

Q2: What genres has highest votes? we have 25.5% of our dataset with popular vote (6520 rows). Drama again gets the highest popularity among fans by being having more than 18.5% of movies popularities.

Q3: What movie got the highest popularity? what's its genre? Spider-Man: No Way Home has the highest popularity rate in our dataset and it has genres of Action, Adventure and Science Fiction.

Q4: What movie got the lowest popularity? what's its genre? The United States of America has the lowest rate in our dataset and it has genres of music, drama, 'war', 'sci-fi' and history.

Q5: Which year has the most filmed movies? year 2020 has the highest filming rate in our dataset.

In []: